Does vibrational strong coupling reorganize the hydrogen-bond network of liquid water? Polarizable machine-learning cavity molecular dynamics

Charlie Wade,[1] **[TODO: co-authors, e.g. model training / experimental collaborators]**, and Johannes Flick[1, 2, 3, *]

[1] *Department of Physics, City College of New York, New York, New York 10031, USA*
[2] *Department of Physics, The Graduate Center, City University of New York, New York, New York 10016, USA*
[3] *Center for Computational Quantum Physics, Flatiron Institute, New York, New York 10010, USA*

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Infrared experiments on liquid water in Fabry–Pérot cavities indicate that vibrational ultra-strong coupling of the O–H stretch slows the thermal disordering of the hydrogen-bond network and redistributes water between network, intermediate and isolated configurations, without any external driving. A microscopic explanation is missing: in classical cavity molecular dynamics with fixed-charge models, the cavity cannot change equilibrium structure at all, so such simulations are blind to this effect by construction. Here we use `cboamd`, a cavity Born–Oppenheimer molecular-dynamics framework that includes the electronic polarizability, the minimal ingredient through which a cavity can bias thermal equilibrium. We first validate the approach on a single CO_2 molecule, reproducing published polariton branches, and on a single water molecule, whose bend polaritons split linearly with coupling. We then follow the collective coupling from one to a few water molecules, establishing **[TODO: $\sqrt{N}$ scaling / separation and orientation dependence]**. Finally, with machine-learned models for the energy, dipole moment and polarizability of liquid water, we simulate the O–H stretch polaritons of the liquid and find that **[TODO: main result: whether and how the hydrogen-bond populations, their temperature dependence and the polariton linewidths change, and by how much compared with experiment]**. **[TODO: One closing sentence on what this means for cavity-modified chemistry in water.]**

I. INTRODUCTION

When molecular vibrations interact strongly with the confined electromagnetic field of an optical cavity, they hybridize with the cavity mode into vibrational polaritons. In this regime of vibrational strong coupling (VSC), the infrared (IR) absorption peak of the molecules splits into a lower and an upper polariton separated by the Rabi splitting, which grows with the square root of the number of coupled molecules [1, 2]. What makes VSC remarkable is that it appears to change matter itself: placing reacting solutions in a resonant infrared cavity, in the dark, has been reported to slow reactions, change their selectivity and modify energy transfer [2–5]. At the same time, careful attempts to reproduce some of these effects found much smaller changes or none at all [6, 7]. Why, and when, a cavity alters the ground state of a molecular ensemble remains an open question.

Water is the natural place to ask this question. It is the solvent of most of chemistry and biology, and its properties are set by a flexible hydrogen-bond network. Its O–H stretch is one of the strongest infrared absorbers, so liquid water reaches the strong- and even ultra-strong-coupling regimes readily. Experiments have reported that collective VSC of water raises its ionic conductivity [8]. Most recently, Itatani *et al.* resolved the O–H stretching band of water under vibrational ultra-strong coupling into contributions from network, intermediate and isolated molecules [9]. They found that the thermal disordering of the hydrogen-bond network is retarded inside the cavity, that the fractions of intermediate and isolated water grow anomalously at elevated temperature, and that the upper polariton narrows towards the homogeneous linewidth of single water molecules. They interpret these observations as vacuum-field-mediated correlations between water molecules that extend beyond those of bulk water. A spectrum alone, however, cannot say which hydrogen bonds change, or why.

Molecular dynamics (MD) simulations can, because they follow every molecule. Cavity MD with empirical force fields has been applied to liquid water under VSC and reproduces the polariton splitting of the O–H stretch [10, 11], as does an extended cavity Born–Oppenheimer treatment [12]. Where the structure of the liquid was examined, it changed little [10, 12]. This is not an accident. For a fixed-charge model, the photon coordinate is a harmonic oscillator whose center depends on the nuclear positions but whose curvature does not; integrating it out of the classical partition function gives a constant, so every equilibrium average of the nuclei is identical inside and outside the cavity [13]. A classical simulation can only see a cavity-induced change of the hydrogen-bond network if the coupling depends on the configuration in a way that goes beyond a fixed dipole. In addition, fixed-charge models misrepresent the light–matter coupling itself: an inaccurate dipole surface directly distorts the Rabi splitting, and for water the widely used q-TIP4P/F model underestimates the O–H stretch intensity and therefore its splitting [11].

The cavity Born–Oppenheimer approximation

* jflick@ccny.cuny.edu

(CBOA) [14] provides a first-principles route that avoids both problems. In its effective electric-field formulation [15], the coupled nuclear–photon dynamics depend only on three field-free properties of the matter: the ground-state energy, the dipole moment and the polarizability. All three can be computed with standard electronic-structure codes and learned by machine-learning interatomic potentials (MLIPs). The resulting framework, `cboamd` [16], makes collective VSC in the condensed phase accessible at near first-principles accuracy. Crucially, the polarizability screens the cavity field in a configuration-dependent way. This renormalizes the effective cavity frequency, the molecular analogue of the refractive index of the intracavity medium, and it is the minimal classical mechanism through which the cavity vacuum can bias thermal equilibrium [16]. For liquid CO_2 the same framework predicts strongly modified IR spectra but an unchanged pair structure [16]. Whether water, with its permanent dipole and cooperative hydrogen bonding, behaves differently is the question we address here.

In this work we [**TODO: one-paragraph summary of what we did and found, written last**]. We build up from the single molecule to the liquid in steps that can each be checked. We first validate our simulations against published results for CO_2 and establish the bend and stretch polaritons of a single water molecule (Sec. IV A). We then couple two to four water molecules to one cavity mode to test collective scaling and to find the separations and orientations at which the molecules couple only through the cavity (Sec. IV B). Finally, we simulate liquid water with machine-learned energy, dipole and polarizability models, compute its O–H stretch polaritons, and analyse the hydrogen-bond network and its temperature dependence in direct comparison with experiment (Secs. IV C and IV D).

II. THEORY

We summarize the effective electric-field formulation of the CBOA [15, 16] for a single cavity mode of frequency ω, coupling strength vector $\boldsymbol{\lambda}$ (atomic units throughout) and photon displacement q. In the dipole approximation and the length gauge, the nuclei with coordinates $\mathbf{R}$ and the photon move on the potential energy surface

$$E(\mathbf{R}, q) = E_e(\mathbf{R}, 0) + \frac{1}{2}\,\frac{\left(\omega q - \boldsymbol{\lambda}\cdot\boldsymbol{\mu}_0(\mathbf{R})\right)^2}{1 + \boldsymbol{\lambda}\cdot\boldsymbol{\chi}_0(\mathbf{R})\cdot\boldsymbol{\lambda}}, \qquad (1)$$

where $E_e(\mathbf{R}, 0)$, $\boldsymbol{\mu}_0(\mathbf{R})$ and $\boldsymbol{\chi}_0(\mathbf{R})$ are the ground-state energy, dipole moment and polarizability of the matter in zero field. The effective field acting on the matter is $\boldsymbol{\mathcal{E}} = \boldsymbol{\lambda}\,\mathcal{E}$ with

$$\mathcal{E} = \frac{\omega q - \boldsymbol{\lambda}\cdot\boldsymbol{\mu}_0(\mathbf{R})}{1 + \boldsymbol{\lambda}\cdot\boldsymbol{\chi}_0(\mathbf{R})\cdot\boldsymbol{\lambda}}, \qquad (2)$$

and the forces on nucleus I and on the photon are

$$\mathbf{F}_I = -\frac{\partial E_e(\mathbf{R}, 0)}{\partial \mathbf{R}_I} + \boldsymbol{\mathcal{E}}\cdot\frac{\partial \boldsymbol{\mu}_0}{\partial \mathbf{R}_I} + \frac{1}{2}\,\boldsymbol{\mathcal{E}}\cdot\frac{\partial \boldsymbol{\chi}_0}{\partial \mathbf{R}_I}\cdot\boldsymbol{\mathcal{E}}, \qquad (3)$$

$$F_q = -\omega\,\mathcal{E}. \qquad (4)$$

The IR spectrum is obtained from the dressed dipole $\boldsymbol{\mu} = \boldsymbol{\mu}_0 + \boldsymbol{\chi}_0\cdot\boldsymbol{\mathcal{E}}$ along the trajectory.

Two consequences of Eq. (1) guide this work. First, the screening term $\boldsymbol{\lambda}\cdot\boldsymbol{\chi}_0\cdot\boldsymbol{\lambda}$ renormalizes the cavity frequency to ω/n_{eff} with $n_{\text{eff}} = (1 + \boldsymbol{\lambda}\cdot\boldsymbol{\chi}_0\cdot\boldsymbol{\lambda})^{1/2}$, the analogue of the refractive index of the intracavity medium [16]. At fixed bare ω this detunes the cavity and makes the Rabi splitting asymmetric; experiments tune the cavity in the presence of the medium, which we mimic with a "resonant" protocol $\omega = n_{\text{eff}}\,\omega_{\text{vib}}$. Second, because $\boldsymbol{\chi}_0(\mathbf{R})$ depends on the configuration, the cavity contribution to the free energy is no longer a constant, so equilibrium structure can in principle change. Neglecting $\boldsymbol{\chi}_0$ (the "χ neglected" dynamics) recovers the fixed-dipole limit, in which it cannot [13]. Comparing the two is therefore a direct test of whether any structural change we find originates in the polarizability.

For N molecules coupled to one mode, the bright collective mode carries the coupling $\Omega_N = \Omega_1\,(\sum_i \cos^2\theta_i)^{1/2}\,Z^*_{\text{env}}/Z^*_1$, where θ_i is the angle between the transition dipole of molecule i and the cavity polarization, and Z^* is the Born effective charge of the coupled vibration, dressed by the environment in the condensed phase. For aligned molecules this reduces to $\sqrt{N}$, and for an isotropic liquid $\sum_i \cos^2\theta_i \to N/3$; this relation connects single-molecule and liquid CO_2 to within a few percent [16]. [**TODO: state the water analogue we test: which mode, which Z^*, and whether polar molecules require a modified model (see Sec. IV A).**]

III. METHODS

A. Cavity molecular dynamics with `cboamd`

All simulations use the public `cboamd` code [16] (commit `89bc6dd`), which propagates Eqs. (3)–(4) with velocity Verlet in the NVE ensemble through an ASE [17] integrator, or through the LAMMPS `fix cboamd` module for condensed phases. We use one cavity mode polarized along the laboratory x axis, a time step of 0.5 fs, and compute IR spectra as the magnitude of the Fourier transform of the x component of the dressed dipole. For every setting we checked, runs made earlier with the group's development version of the code give bit-identical trajectories. Energy conservation is monitored with the full conserved quantity $E_{\text{kin}} + E_e + \frac{1}{2}p^2 + \frac{1}{2}(1 + \boldsymbol{\lambda}\cdot\boldsymbol{\chi}_0\cdot\boldsymbol{\lambda})\,\mathcal{E}^2$, which fluctuates by about 1% with a drift below 0.3% in all single-molecule runs.

B. Single molecules and small clusters

Single molecules and clusters of up to four water molecules are treated with on-the-fly density-functional theory in PySCF [18]. For the exploratory runs reported here we use the local-density approximation (LDA, VWN) with the cc-pVDZ basis set; production runs will use [**TODO: functional and basis, e.g. PBE0/aug-cc-pVDZ as in Ref. 16**]. Dipole derivatives are analytic and polarizability derivatives are obtained by finite differences. Each trajectory starts from the equilibrium geometry with a small displacement along the mode of interest and zero velocities, with the photon at rest at $q = 0$ [**TODO: or at its relaxed value $q_0 = \boldsymbol{\lambda} \cdot \boldsymbol{\mu}_0/\omega$, to be decided; see Sec. IV A**]. The single water molecule is oriented with its C_2 axis along the cavity polarization, so that the bend and symmetric stretch couple to the mode.

For CO_2 we additionally use the NEP models for the energy, dipole and polarizability distributed with `cboamd`, trained on PBE0/aug-cc-pVDZ data [16].

Clusters of $N = 2$–4 water molecules are built from aligned copies of the single molecule at center-of-mass separation R. [**TODO: arrangement (head-to-tail along x vs. side by side), separations scanned (planned 4–10 Å), the criterion for "cavity-only" coupling, and how dark modes are excited.**] [**TODO: whether molecular rotation is allowed or constrained.**]

C. Machine-learned models for liquid water

[**TODO: Choose: reuse published deep-learning models for the dipole and polarizability of liquid water [19, 20] or train new ones. Describe the reference electronic structure (functional, basis or plane-wave cutoff, dispersion), the source of training configurations (e.g. AIMD or MLIP sampling at several temperatures, with and without the cavity), the energy/force model, the dipole model (molecular dipoles from Wannier centers), the polarizability model, the training framework (DeePMD-kit [21] or NEP), and the test errors (table).**]

D. Liquid-water simulations

[**TODO: System size (number of molecules, box length, density), equilibration (thermostat, temperature range to match the experiment), production protocol (number and length of independent NVE trajectories per setting), cavity frequency (resonant with the liquid O–H stretch, and possibly the bend), coupling strengths and how they map onto the experimental Rabi splitting, and the three protocols: χ neglected, χ included at fixed ω, and χ included with $\omega = n_{\mathrm{eff}}\,\omega_{\mathrm{OH}}$.**]

E. Analysis

[**TODO: IR spectra and polariton peak positions and linewidths (averaging, smoothing, fits); decomposition of the O–H band; classification of molecules as network, intermediate and isolated by their number of hydrogen bonds, with the geometric hydrogen-bond criterion used, chosen to mirror the experimental decomposition of Ref. 9; populations versus temperature; radial distribution functions, hydrogen-bond lifetimes and orientational correlation as structural and dynamical observables; statistical error estimates.**]

IV. RESULTS

A. Validation and single molecules

We first reproduced the single-CO_2 benchmark of Ref. 16. With the cavity resonant with the asymmetric stretch, the polariton branches and Rabi splittings from our NEP-driven trajectories coincide with the published benchmark at every coupling strength, both with the polarizability included and neglected (Fig. 1); at $\lambda = 0.1$ with χ included, we obtain the lower and upper polaritons at 2105 and 2525 cm^{-1} ($\Omega_R = 420$ cm^{-1}). Direct PySCF dynamics agree with the NEP results within their frequency resolution. Including the polarizability raises the splitting by about 15% and brings it within about 5% of the explicit QEDFT reference [15]. A detuning scan follows the linear Hopfield model including the dipole self-energy to within 3 cm^{-1}, and the splitting is independent of the initial displacement, confirming that the dynamics are in the linear regime.

For a single water molecule with the cavity resonant with the bend, the bend splits into two polaritons whose separation grows linearly with coupling, $\Omega_R \approx 1265\,\lambda$ cm^{-1} with the polarizability neglected (Fig. 2). With the polarizability included ($\chi_{xx} \approx 5.7$ a.u.), one polariton dominates and tracks the screened cavity frequency ω/n_{eff}, as for CO_2. Two features distinguish the polar water molecule from CO_2. When the cavity polarization is tilted away from its dipole, the molecule rotates appreciably, and a detuning scan lies 40–60 cm^{-1} below the linear model that describes CO_2. [**TODO: resolve: origin of the offset (photon initial condition, permanent dipole, anharmonicity); repeat for the O–H stretch; resonant-protocol runs with χ included; longer runs.**]

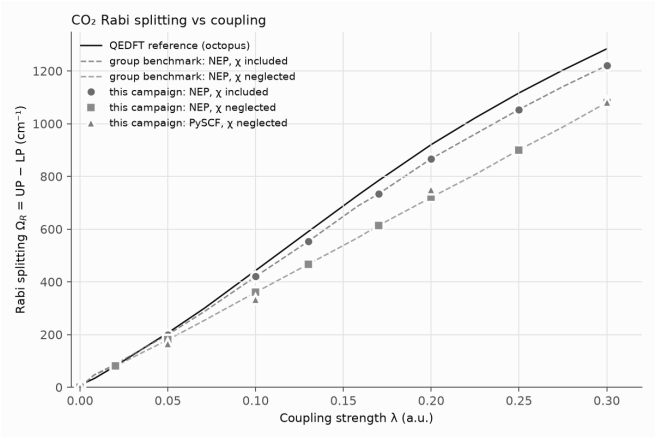


FIG. 1. Validation on a single CO_2 molecule: Rabi splitting versus coupling strength. Circles and squares, this work with NEP models with the polarizability included and neglected; triangles, PySCF with the polarizability neglected; dashed lines, benchmark of Ref. 16; solid line, explicit CBOA QEDFT reference [15]. **[TODO: redraw in final style; LDA vs. PBE0 for the PySCF points.]**

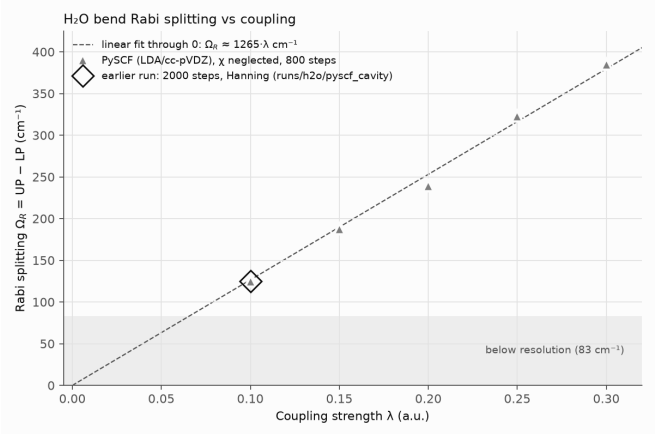


FIG. 2. Single water molecule (PySCF, LDA/cc-pVDZ, polarizability neglected) with the cavity resonant with the bend: Rabi splitting versus coupling strength, with a linear fit through the origin. The shaded band marks the frequency resolution of the 800-step trajectories. **[TODO: replace with production level of theory and include the O–H stretch.]**

B. From one to a few molecules

[TODO: Separation scan for $N = 2$: bare spectra and splitting versus R; the distance beyond which molecules couple only through the cavity. Collective scaling for $N = 1$–4 at fixed λ (expected $\Omega_R \propto \sqrt{N}$) and at $\lambda/\sqrt{N}$ (expected constant); role of orientation and rotation; dark modes.]

Planned figure: Rabi splitting versus N (1–4) at fixed λ and at $\lambda/\sqrt{N}$, with the $\sqrt{N}$ prediction; inset, splitting versus separation R for $N = 2$.

FIG. 3. **[TODO: Collective scaling in small water clusters.]**

Planned figure: IR spectra of liquid water in the O–H stretch region versus coupling: χ neglected, χ included at fixed ω, and resonant protocol.

FIG. 4. **[TODO: Polaritons of liquid water.]**

C. Liquid water: O–H stretch polaritons

[TODO: IR spectra of liquid water versus coupling for the three protocols; Rabi splitting versus λ and its connection to the single-molecule value through orientation and local-field factors; comparison with the experimental splitting; upper-polariton linewidth versus coupling, compared with the narrowing reported in Ref. 9.]

D. Hydrogen-bond network

[TODO: Populations of network, intermediate and isolated water inside and outside the cavity and their temperature dependence; whether the polarizable dynamics show the retarded disordering seen experimentally while the χ-neglected dynamics do not; RDFs, hydrogen-bond lifetimes and correlation lengths; size of any effect relative to statistical error.]

V. DISCUSSION

[TODO: Interpret the results against the experiment of Ref. 9: does the polarizability-

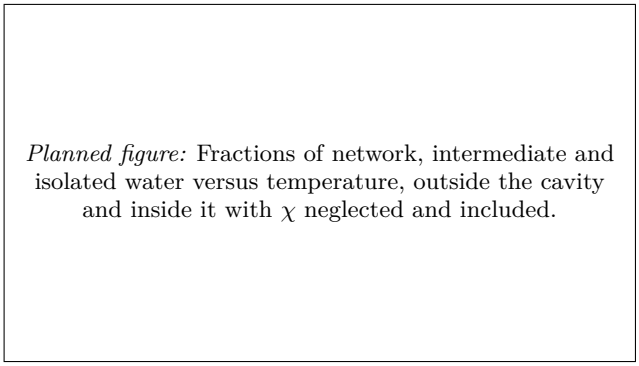


FIG. 5. **[TODO: Hydrogen-bond populations under VSC.]**

mediated mechanism account for the observed network reorganization, and at what coupling strength; what classical, mean-field CBOA dynamics can and cannot capture (nuclear quantum effects, many cavity modes, cavity losses, realistic molecule numbers); implications for cavity-modified chemistry in aqueous solution.]

VI. CONCLUSIONS

[TODO: Three or four sentences: what we did, the main finding, and the outlook.]

ACKNOWLEDGMENTS

[TODO: Funding, computing resources (e.g. Flatiron Institute), and discussions.]

DATA AVAILABILITY

Inputs, analysis scripts and data for all simulations are available at `https://github.com/charliew0707/VSC_Molecular_Dynamics` **[TODO: archive with a DOI at submission]**; the simulation code is available at `https://github.com/flickgroup/cboamd`.

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